

Tensores

$$\vec{V} = \underbrace{v^i}_{\text{comp. contravariantes}} \underbrace{\vec{e}_i}_{\text{comp. covariantes}} = v_i \vec{e}^i$$

$$q^i \rightarrow q^{i'}$$

$$V^{i'} = \frac{\partial q^{i'}}{\partial q^k} V^k \quad \text{e} \quad V_{i'} = \frac{\partial q^k}{\partial q^{i'}} V_k$$

$$T^{ij} \rightarrow (2,0) \quad q^i \rightarrow q^{i'} \quad T^{i'j'} = \frac{\partial q^{i'}}{\partial q^k} \frac{\partial q^{j'}}{\partial q^m} T^{km}$$

$$g_{ij} = \vec{e}_i \cdot \vec{e}_j \quad (0,2) \quad \text{e} \quad g^{ii} = \vec{e}^i \cdot \vec{e}^i \quad (2,0)$$

$$\text{e p/ } T^i_j \xrightarrow{q \rightarrow q'} T^{i'}_{j'} = \frac{\partial q^{i'}}{\partial q^k} \frac{\partial q^l}{\partial q^{j'}} T^k_l$$

ainda:

$$T^{i \quad k}_{j \quad m} (2,2) \rightarrow T^{i' \quad k'}_{j' \quad m'} = \frac{\partial q^{i'}}{\partial q^r} \frac{\partial q^s}{\partial q^{j'}} \frac{\partial q^{k'}}{\partial q^l} \frac{\partial q^t}{\partial q^{m'}} T^{st}_{rl}$$

Tensor de Riemann

Proposição: a derivada covariante de um vetor é um tensor

$$\vec{V} = v^i \vec{e}_i \quad \frac{\partial V}{\partial q^i} = \underbrace{\left(\frac{\partial v^k}{\partial q^i} \right)}_{V^k_{;i}} \vec{e}_k$$

Queremos mostrar que mediante a:

$$q^i \rightarrow q^{i'} \quad V^{k'}_{i'} = \frac{\partial q^{k'}}{\partial q^l} \frac{\partial q^r}{\partial q^{i'}} V^l_r$$

$$\frac{\partial V}{\partial q^i} = \frac{\partial V}{\partial q^{k'}} \frac{\partial q^{k'}}{\partial q^i} = V^{m'}_{jk'} e_{m'} \frac{\partial q^{k'}}{\partial q^i} \quad \text{e} \quad \vec{e}_{m'} = \frac{\partial \vec{S}}{\partial q^{m'}} = \frac{\partial \vec{S}}{\partial q^r} \frac{\partial q^r}{\partial q^{m'}} \quad \vec{e}_r$$

$$\underline{V^r_{ji}} \vec{e}_r = V^{m'}_{jk'} \underbrace{\frac{\partial q^r}{\partial q^{m'}} \frac{\partial q^{k'}}{\partial q^i}}_{\vec{e}_{m'}} \vec{e}_r$$

$$\vec{e}_{m'} = \frac{\partial q^r}{\partial q^{m'}} \vec{e}_r$$

$$V^r_{ji} = \frac{\partial q^r}{\partial q^{m'}} \frac{\partial q^{k'}}{\partial q^i} V^{m'}_{jk'}$$